

P is the set of all polynomial decidable languages.
(polynomial) $\rightarrow$ DTM

Theorem

All context-free languages are in P .

Proof.

Let L be any context-free language.

$\exists$ CFG $G = (V, \Sigma, S, R)$ in CNF that generates L .

M : on input w to decide $w \in L$ or not.

$$w = a_1 \dots a_n$$

Subproblem: for $1 \leq i \leq j \leq n$, define $T[i, j]$ be the set
 $\{A \in V - \Sigma : A \Rightarrow_G^* a_i \dots a_j\}$

Goal: $T[1, n]$ $S \in T[1, n]$?

$$S \Rightarrow_G^* a_1 \dots a_n = w$$
$$w \in L$$

Base case: for $1 \leq i \leq n$

$$T[i, i] = \{A \in V - \Sigma : A \Rightarrow_G^* a_i\}$$

$$= \{A \in V - \Sigma : A \rightarrow a_i \text{ is a rule}\}$$

Recurrence: for $1 \leq i < j \leq n$

$$T[i, j] = \bigcup_{k=i}^{j-1} \{A \in V - \Sigma : A \rightarrow BC, B \in T[i, k], C \in T[k+1, j]\}$$

for some k such that $i \leq k < j$

$$B \Rightarrow_G^* a_i \dots a_k \quad \text{i.e. } B \in T[i, k]$$

$$C \Rightarrow_G^* a_{k+1} \dots a_j \quad \text{i.e. } C \in T[k+1, j]$$

$$A \rightarrow BC$$



$$A \Rightarrow_G^* a_i \dots a_j \quad A \in T[i, j]$$

$$T[i, j] \text{ s.t. } j-i=0$$

$$T[i, j] \text{ s.t. } j-i=1$$

$$\vdots \quad j-i=2$$

M : on input w

1. for $i=1, \dots, n$ // $n=|w|$

2. compute $T[i, i]$

3. for $\ell=1, \dots, n-1$

4. for $i=1, \dots, n-\ell$

5. $j=i+\ell$

6. $T[i, j] = \bigcup_{k=i}^{j-1} \{ \dots \}$

7. if $S \in T[1, n]$



$n \cdot |R|$

$O(n^2)$

$\cdot O(n|R|)$

8. accept w

9. else

10. reject

for $k = i$ to $j-1$

for each rule $A \rightarrow BC$

if $B \in T[i, k]$

and $C \in T[k+1, j]$

put A in $T[i, j]$

$O(n^3 \cdot |R|)$

$O(n^3)$

$L \in P$

SAT (satisfiability) Problem

Let $X = \{x_1, x_2, \dots, x_n\}$ be a set of boolean variables.

$x_1, \dots, x_n, \bar{x}_1, \dots, \bar{x}_n$ = Literals

a clause is a set of literals $\{x_1, \bar{x}_2, x_4\}$ $x_1 \vee \bar{x}_2 \vee x_4$

a boolean formula (in conjunctive normal form) is a set of clauses

$(x_1 \vee \bar{x}_2 \vee x_4) \wedge (\bar{x}_1 \vee x_3) \wedge (x_2 \vee \bar{x}_3 \vee x_4)$

A truth assignment T for X is a mapping from X to $\{0, 1\}$

A truth assignment $\overset{T}{\text{satisfies}}$ a boolean formula $\bar{F}$ if

for each clause $C \in \bar{F}$, there is at least one boolean variable x_i ,

either $x_i \in C$ and $T(x_i) = 1$

or $\bar{x}_i \in C$ and $T(x_i) = 0$

is satisfiable

SAT:

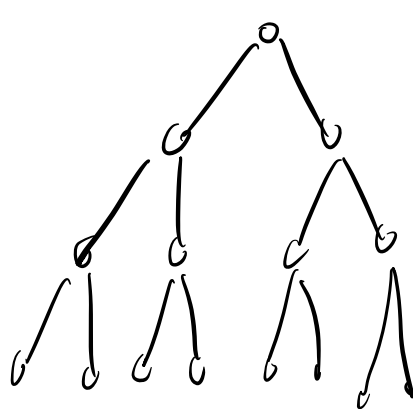
Given a boolean formula F , is F satisfiable?

$\{ "F" : F \text{ is a satisfiable boolean formula} \}$

$SAT \in P$? unknown

Let N be a non-deterministic TM such for any input, every branch of non-determinism halts.

The running time of N is a function $f: \mathbb{N} \rightarrow \mathbb{N}$ s.t. $f(n)$ is the maximum steps used by any branch of non-determinism on any input of size n .



height
= running time.

Theorem

SAT is decided by some NTM that runs in time $\text{poly}(n)$.

Proof:

M : on input $"F"$

1. non-deterministically generate a truth assignment T
2. If T satisfies $\bar{F}$
3. accept
4. else
5. reject

NP is the set of all languages that are polynomial-time decidable by NTM.
(non-deterministically polynomial)

$$\text{SAT} \in \text{NP}$$
$$P \subseteq \text{NP}$$

A language L is polynomial-time verifiable if there is a polynomial-time DTM V such that for any string x

$x \in L$ if and only if for some string c of size at most $\text{poly}(|x|)$ ^{certificate}

V accepts " x " " c ".
(
verifier

SAT is polynomial-time verifiable.

for any satisfiable F , let c be the truth assignment satisfying F .

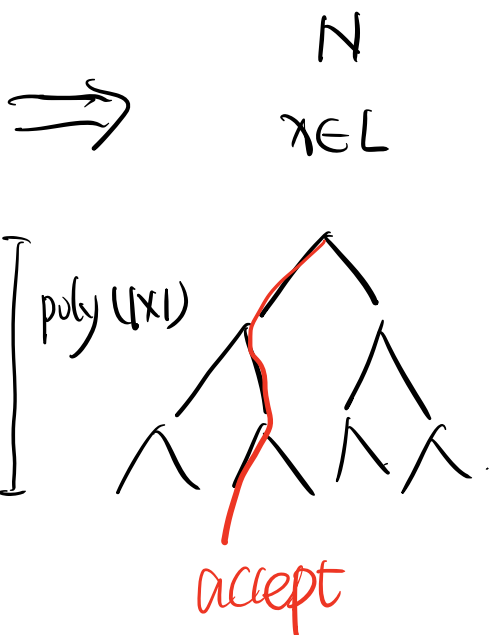
$V =$ on input " $\bar{F}$ " " c "

1. evaluate F under c .
2. if F is 1
3. accept.
4. else
5. reject.

Theorem $\overset{L}{\underset{V}{\hookrightarrow}}$

A language $\overset{L}{\underset{V}{\hookrightarrow}}$ is in NP iff it is polynomial-time verifiable.

Proof:



$x \in L$ of N
 $\checkmark$
 c is the branch that accept x

$V =$ on input " x " " c "

1. run N on x under the guidance of c
2. if N accept c
3. accept " x " " c "
4. else
5. reject

$\Leftarrow$)

N : on input x

1. non-deterministically generate c whose size is at most $\text{poly}(|x|)$

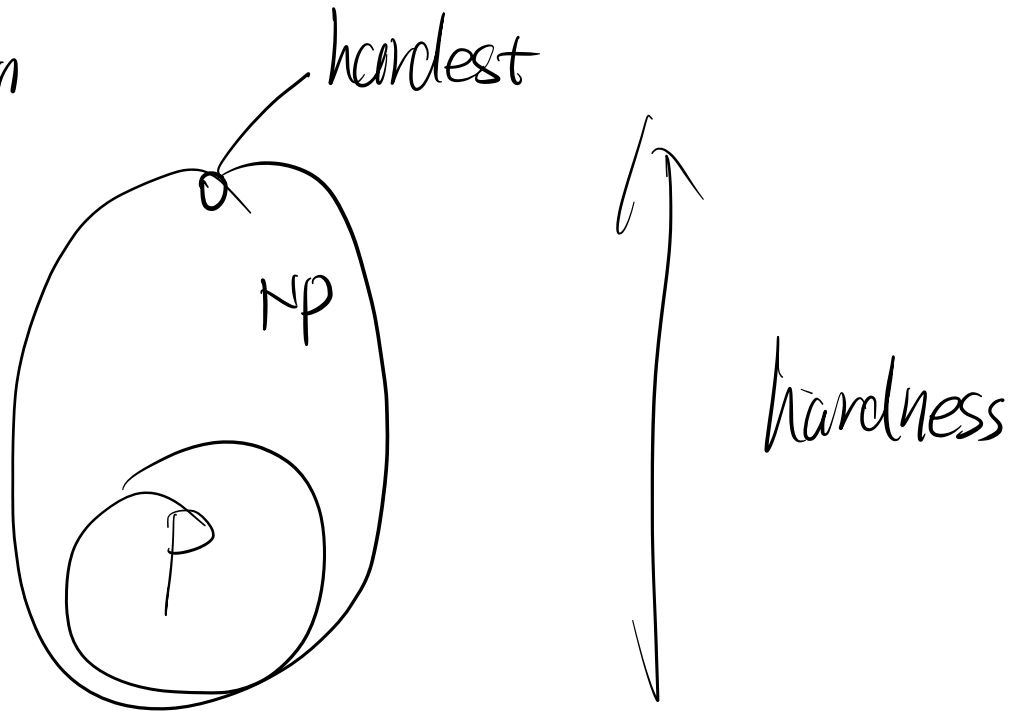
for $x \in L$
 $\exists c$

V accepts " x " " c "

2. run V on " x " " c "

3. If V accepts " x " " c " accept x

$P \subseteq NP$
 $P = NP?$ unknown



$\downarrow$
A reduction from A to B is a polynomial-time reduction if for any x , $f(x)$ is computed by a DTM

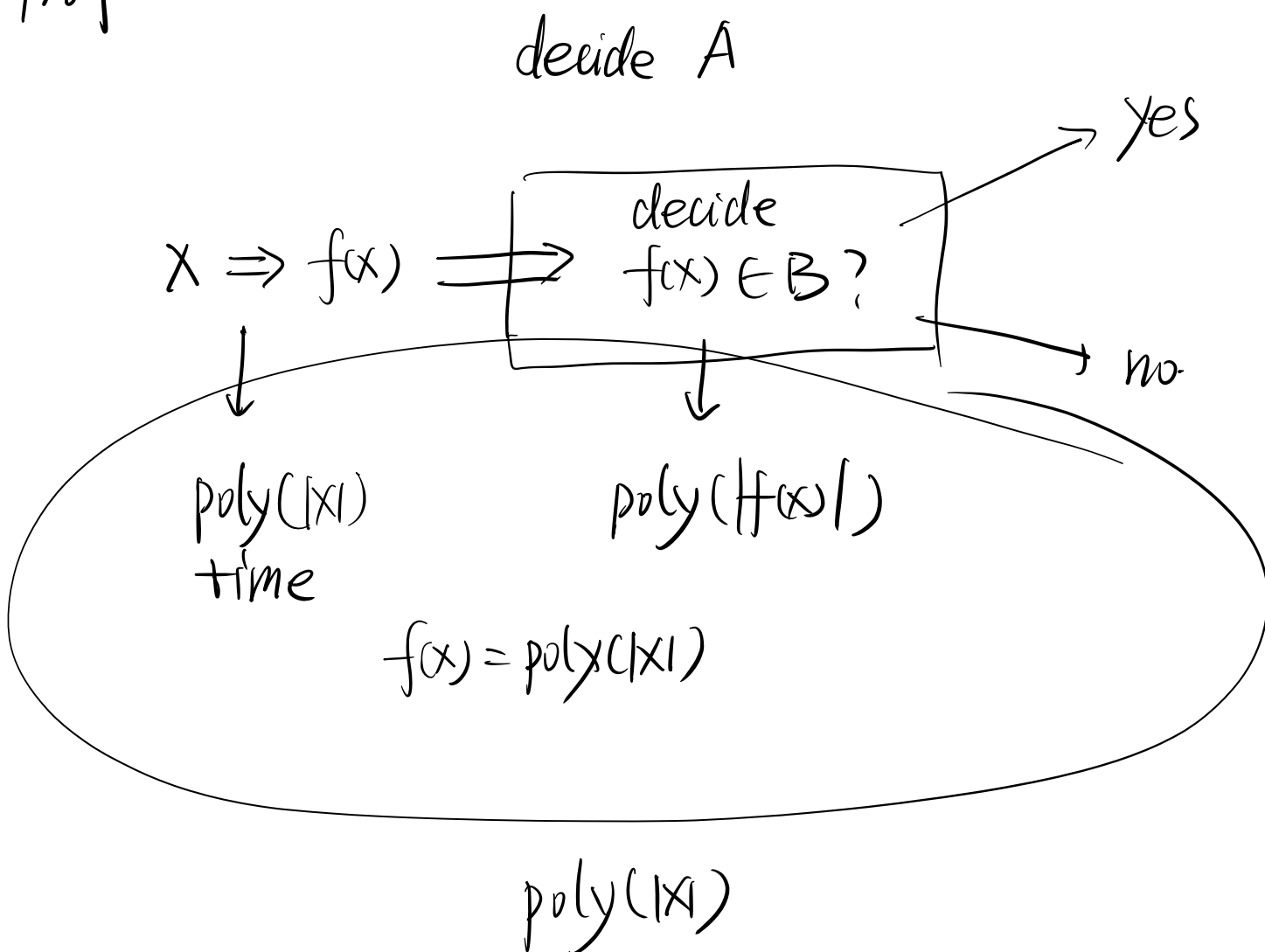
in $\text{poly}(|x|)$.

$$A \leq_p B$$

Theorem

If $A \leq_p B$ and $B \in P$, then $A \in P$.

Proof:



Definition

A language L is NP-complete if

(1) $L \in NP$

(2) any language in NP can be reduced to L
in polynomial-time.

Corollary

If a NP-complete language is in P, then $NP \subseteq P$
then $NP = P$.